Fibonacci Numbers

Time limit: 1.00 s
Memory limit: 512 MB

CSES 1722 | Mathematics | https://cses.fi/problemset/task/1722

The Fibonacci numbers can be defined as follows:

- $F_0=0$
- $F_1=1$
- $F_n = F_{n-2}+F_{n-1}$

Your task is to calculate the value of F_n for a given n.

Input

The only input line has an integer n.

Output

Print the value of F_n modulo 10^9+7.

Constraints

- $0 \le n \le 10^{18}$

Example

Input
10

Output
55